

Fast, Quantization Aware DNN Training for Efficient HW Implementation

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Abstract—Quantization of Deep Neural Networks is a central technique to reduce the computation load in embedded devices. Even in quantized Deep Neural Networks (DNNs), the scaler/rescaler following a convolution or dense layer often requires a high bit width multiplication and a shift. Previous work has proposed to remove the multiplier by restricting the quantization method. We propose a Quantisation Aware Training (QAT) approach, which explicitly models the rescaler during training, eliminating the limitations of quantization functions and achieving a 30 – 35% improvement in training time and a significant reduction in memory requirements compared to the state-of-the-art. [GitHub:https://github.com/embedded-machine-learning/FastQATforPOTRescaler](https://github.com/embedded-machine-learning/FastQATforPOTRescaler)

Index Terms—Convolution, Training, Neural networks, Quantization (signal), Quantization Aware Training, hardware-friendly

I. INTRODUCTION

Due to resource constraints, inference of deep neural networks on edge devices remains challenging. Various optimization approaches have been proposed, including approximate matrix multiplications, pruning, and quantization. While pruning focuses on reducing the model size by removing unnecessary connections and weights, approximate matrix multiplications aim to perform computationally expensive operations more efficiently. On the other hand, quantization offers a dual benefit by reducing the model size while enabling more efficient hardware utilization. For example, NVIDIA claims up to $4\times$ acceleration when transitioning from FP32 to INT8¹. Using a Field programmable Gate Array (FPGA) or Application-Specific Integrated Circuits (ASIC) could further improve the performance, particularly because the size of a standard multiplier scales with the squared number of bits.

A. Operations of an Accelerator

Dense/Convolutional Layers require matrix multiplications, which in turn require Multiply Accumulate (MAC) operations (Fig. 1). A MAC operation is part of a Digital Signal Processor (DSP). The MAC is characterized by the number of bits for the weights and the feature maps. For instance, a common notation

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¹Accessed: 10th of May 2023 <https://developer.nvidia.com/blog/nvidia-ampere-architecture-in-depth/>

is A8W4, which indicates that the activation function produces an 8-bit value (typically the feature map for the following input), and the weights are represented in a 4-bit format. The Accumulation register is typically implemented using a 32-bit data type or is assumed to be large enough to avoid overflow.

FPGAs often have DSPs. These units have predefined sizes. For example, the Xilinx ultra-96 has 350 28×18 -bit multiplier with a 48-bit accumulation register². Since these DSPs are explicitly implemented in hardware, they can not be changed. However, they can be used in a modified manner, e.g., more energy efficient or multiple multiplications at once.

The matrices in Deep Neural Networks (DNNs) are often quite large. Therefore, matrix multiplications require numerous multiplications, making them computationally demanding. Minimizing the number of multiplications in other parts of the network can significantly enhance the overall performance. Other layers within DNNs that rely on multiplication operations include the Batch Normalization (BN)-Activation functions, which result in rescaling operations in hardware. Hence, reducing multiplications in these layers can improve the overall inference speed in DNNs.

B. Rescaler

The role of the Rescaler in DNN accelerators is to reduce the accumulation register to match the bit-width of the feature

²Accessed: 3rd of May 2023 <https://docs.xilinx.com/v/u/en-US/ug579-ultrascale-dsp>

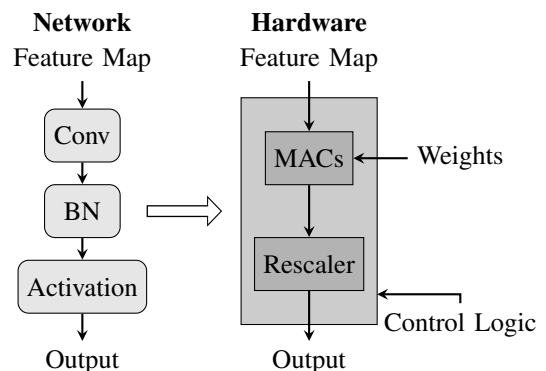


Fig. 1. Network operations and the HW Blocks for their implementation. (Conv=Convolution, BN=Batch Normalization, MAC=Multiply-Accumulate Unit)